

Name: _____

Maths Class Teacher: _____

SYDNEY TECHNICAL HIGH SCHOOL



Extension 2 Mathematics

HSC Assessment Task 1

March 2011

General Instructions

- Working time – 70 minutes
- Write using **black or blue pen**
- Board-approved calculators may be used
- **All necessary working** should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Start each question on a **new page**.
- Place your papers **in order** with the question paper on top and staple or pin them.

Total Marks - 50

- Attempt Questions 1 – 3
- Mark values are shown with the questions.

(For markers use only)

Q1	Q2	Q3	Total
17	16	17	50

Question 1**17 Marks**

- (a) In each case below. $z = 3 - 2i$. 4

Express the following in the form $x + iy$ where x and y are real numbers:

(i) $\overline{(iz)}$

(ii) $(z - 1)^2$

Evaluate:

(iii) $\arg(z)$ (in radians to one decimal place)

(iv) $|z|$ (leave in exact form)

- (b) (i) Express $1 + i\sqrt{3}$ in modulus argument form. 2

(ii) Hence evaluate $(1 + i\sqrt{3})^5$ 2

(Express your answer in the domain defined for the argument.)

- (c) For $z = x + iy$,

(i) Express $\frac{1}{z}$ as a complex number. 1

(ii) Hence find the solutions for z if $\operatorname{Re}\left(z - \frac{1}{z}\right) = 0$, 3

and show on the Argand diagram.

- (d) Sketch the following loci.

(i) $|\arg z| \leq \frac{\pi}{4}$ 1

(ii) $|z + 2| + |z - 2| = 6$ 1

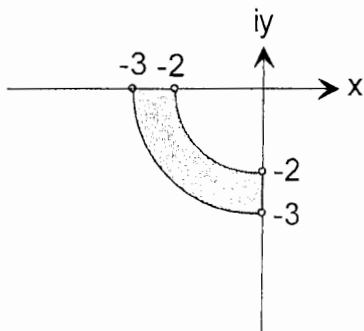
- (e) Sketch the region where the inequalities $|z - 2 + i| \leq 5$ and $|z - 1| \geq |z + 1|$ both hold. 3

Question 2

16 Marks

- (a) Give the inequalities which describe this region on the Argand diagram.
(Give your answer in terms of z .)

3



- (b) If $1 - 2i$ is a root of the equation $2z^3 - 5z^2 + 12z - 5 = 0$

(i) Explain why $1 + 2i$ is also a root.

1

(ii) Find all roots of the equation.

2

- (c) (i) Show on an Argand diagram the positions of the roots of $z^3 = -1$.

1

(ii) Explain algebraically why these are among the roots of $z^6 = 1$.

2

(iii) By referring to the roots of $z^6 = 1$, find the roots of $z^4 + z^2 + 1 = 0$ in mod-arg form.

3

- (d) (i) Solve $z^4 = 1$ for all z .

1

(ii) Hence, or otherwise, solve $z^4 = (z - 1)^4$.

3

Question 3**17 Marks**

- (a) Sketch the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$. State the following: **5**
- (i) the eccentricity
 - (ii) the coordinates of the foci
 - (iii) the equations of the directrices.
- (b) Find the equation of the tangent to the curve $x^2 - xy^2 - 8y + 32 = 0$ at the point $(1, 3)$. **3**
- (c) Prove that the equation of the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(x_1, y_1)$ can be expressed as $\frac{x_1 y}{a^2} - \frac{x y_1}{b^2} = \frac{x_1 y_1}{a^2} - \frac{x_1 y_1}{b^2}$. **4**
- (d) The tangent to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(x_1, y_1)$ cuts the y axis at A while the normal at P cuts the y axis at B. If S is a focus of the ellipse, show that $\angle ASB = 90^\circ$. (Equation of tangent $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$) **5**

End of Exam

SOLUTIONS EXTENSION 2 ASSESSMENT 1 2011

1 a) $z = 3 - 2i$

(i) $iz = 3i + 2$
 $= 2 + 3i$

$\therefore \bar{iz} = 2 - 3i$

(ii) $\arg z = \tan^{-1}\left(\frac{-2}{3}\right)$
 $\approx -33.69^\circ$
 or -0.6°

(iv) $|z| = \sqrt{3^2 + (-2)^2}$
 $= \sqrt{9 + 4}$
 $= \sqrt{13}$

(ii) $(z-1)^2 = (3-2i-1)^2$
 $= (2-2i)^2$
 $= 4 - 4 - 8i$
 $= -8i$

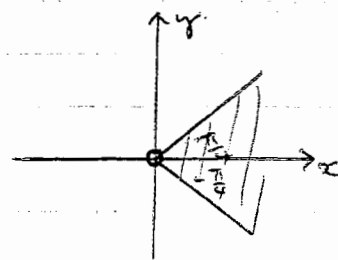
b)(i) $z = 1 + i\sqrt{3}$
 $|z| = \sqrt{1+3}$
 $= 2$

b(ii) $(1+i\sqrt{3})^5 = (2 \operatorname{cis} \frac{\pi}{3})^5$
 $= 32 \operatorname{cis} \frac{5\pi}{3}$
 $= 32 \operatorname{cis} \left(\frac{5\pi}{3}\right)$
 $\arg z = \tan^{-1}(\sqrt{3})$
 $= \frac{\pi}{3}$

$\therefore z = 2 \operatorname{cis} \frac{\pi}{3}$

c)(i) $\frac{1}{z} = \frac{1}{x+iy} \times \frac{x-iy}{x-iy}$
 $= \frac{x-iy}{x^2+y^2}$

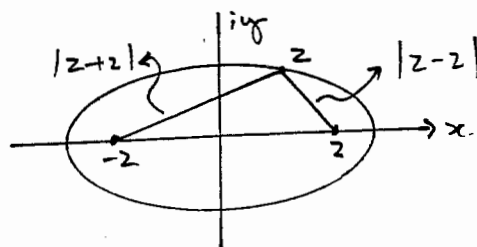
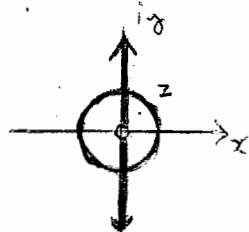
d)(i) $|\arg z| \leq \frac{\pi}{4}$



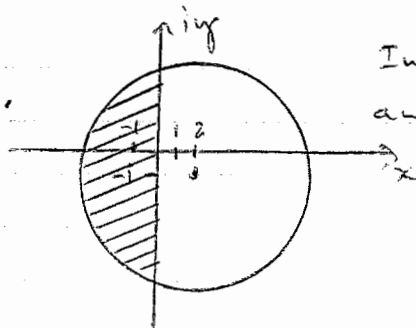
(ii) $\operatorname{Re}\left(z - \frac{1}{z}\right) = \operatorname{Re}\left[(x+iy) - \frac{x-iy}{x^2+y^2}\right], z \neq 0$
 $= x - \frac{x}{x^2+y^2}$

(ii) $|z+2| + |z-2| = 6$

$\therefore x\left(1 - \frac{1}{x^2+y^2}\right) = 0$
 $\therefore x=0$ or $x^2+y^2=1$



e) $|z-2+i| \leq 5$ and $|z-1| \leq |z+1|$



Inside circle centre $2-i$ with radius 5 (including circle) and left of y -axis.

Q2

a) $2 \leq |z| \leq 3$

and $-\pi \leq \arg z \leq -\frac{\pi}{2}$

(i) $z^4 = 1$

$\therefore z = \pm 1, \pm i$

(ii) $z^4 = (z-1)^4$

$\therefore \frac{z^4}{(z-1)^4} = 1$

$\therefore \left(\frac{z}{z-1}\right)^4 = 1$

$\therefore \frac{z}{z-1} = \pm 1, \pm i$

For $\frac{z}{z-1} = 1$

$z = z-1$ no solution

For $\frac{z}{z-1} = -1$

$z-1$

$z = 1-z$

$\therefore z = \frac{1}{2}$

For $\frac{z}{z-1} = i$

$z-1$

$z = i(z-1)$

$z - iz = -i$

$z(1-i) = -i$

$\therefore z = \frac{-i}{1-i}$

For $\frac{z}{z-1} = -i$

$z-1$

$z = -i(z-1)$

$\therefore z + iz = 1$

$\therefore z(1+i) = 1$

$\therefore z = \frac{1}{1+i}$

$\therefore z = \frac{1}{2}, \frac{-1}{1-i}, \frac{1}{1+i}$

b) $2z^3 - 5z^2 + 12z - 5 = 0$

(i) The coefficients are real
 $\therefore$ complex roots consist of conjugates. $\therefore$ If $1-2i$ is a root then $1+2i$ is also a root.

(ii) $(z - (1-2i))(z - (1+2i))$

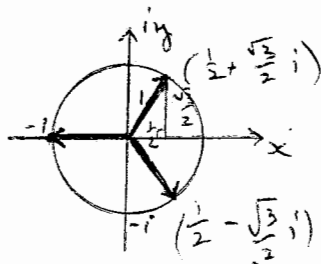
$= z^2 - (1-2i)z - (1+2i)z + (1-2i)(1+2i)$

$= z^2 - 2z + 5$

$2z^3 - 5z^2 + 12z - 5 = (z^2 - 2z + 5)(2z - 1)$

$\therefore$ roots are $z = \frac{1}{2}, 1-2i, 1+2i$

(i) $z^3 = -1$



(ii) $z^6 = 1$

$\therefore z^6 - 1 = 0$

$\therefore (z^3 + 1)(z^3 - 1) = 0$

$\therefore z^3 = 1 \text{ or } -1$

 $\therefore$ the 6 roots of $z^6 = 1$ include the 3 roots of $z^3 = 1$

(iii) $z^6 = 1$ has 1 and -1 as roots.

Now $z^6 - 1 = 0$

$\therefore z^6 - 1 = (z+1)(z-1)(z^4 + z^2 + 1)$ — A

Let $z = \cos \theta$

$\therefore (\cos \theta)^6 = 1$

$\therefore \cos 6\theta = 1$

Now $\cos(2\pi k) = 1$ where $k = 0, 1, \dots, 5$

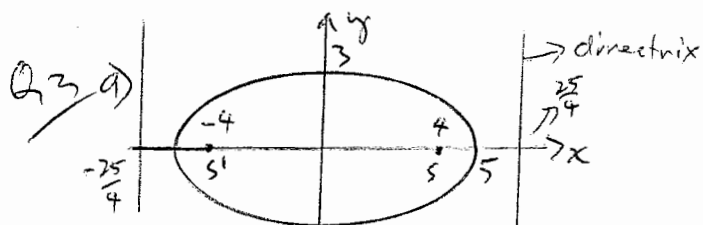
$\therefore z = (\cos 2\pi k)^{\frac{1}{6}} = 1$ for $k = 0, \dots, 5$

$= \cos\left(\frac{\pi k}{3}\right)$

$= 1, \cos \frac{\pi}{3}, \cos \frac{2\pi}{3}, \cos \frac{3\pi}{3} = -1, \cos \frac{4\pi}{3}$

and $\cos \frac{5\pi}{3}$

In A the factors $z+1$ and $z-1$ use the roots ± 1 $\therefore$ the roots of $z^4 + z^2 + 1$ are $\cos \frac{\pi}{3}, \cos \frac{2\pi}{3}, \cos \frac{4\pi}{3}, \cos \frac{5\pi}{3}$



$$a = 5, b = 3$$

$$a^2 e^2 = a^2 - b^2$$

$$= 25 - 9$$

$$= 16$$

$$1 - e^2 = \frac{b^2}{a^2}$$

$$= \frac{9}{25}$$

ii) $\therefore ae = 4 \rightarrow \text{focus } (4, 0)$ $\therefore e^2 = 1 - \frac{9}{25}$

$$= \frac{16}{25}$$

(i) $\therefore e = \frac{4}{5} \rightarrow \text{eccentricity}$

b) $x^2 - xy^2 - 8y = 0$ (1, 3)

$$\therefore 2x - y^2 - 2xy \cdot \frac{dy}{dx} - 8 \frac{dy}{dx} = 0$$

$$\therefore 2x - y^2 - \frac{dy}{dx} (2xy + 8) = 0$$

$$\therefore \frac{dy}{dx} = \frac{2x - y^2}{2xy + 8}$$

$$= \frac{2 - 9}{6 + 8} \text{ at } (1, 3)$$

$$= \frac{-7}{14}$$

$$= -\frac{1}{2}$$

eqn of tangent $\Rightarrow y - y_1 = m(x - x_1)$

$$y - 3 = -\frac{1}{2}(x - 1)$$

$$2y - 6 = -x + 1$$

$$\therefore x + 2y - 7 = 0$$

c) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\therefore \frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$\therefore$ slope of normal at (x_1, y_1)

$$= \frac{a^2 y_1}{b^2 x_1}$$

$\therefore$ eqn of normal $\Rightarrow$

$$y - y_1 = \frac{a^2 y_1}{b^2 x_1} (x - x_1)$$

$$\therefore b^2 x_1 y - b^2 x_1 y_1 = a^2 x y_1 - a^2 x_1 y_1$$

$$\therefore \frac{x_1 y}{a^2} - \frac{x_1 y_1}{a^2} = \frac{x y_1}{b^2} - \frac{x_1 y_1}{b^2}$$

$$\therefore \frac{x_1 y}{a^2} - \frac{x y_1}{b^2} = \frac{x_1 y_1}{a^2} - \frac{x_1 y_1}{b^2}$$

QED

$$\therefore \text{slope AS} = \frac{b^2}{y_1} \quad \text{slope BS} = \frac{y_1 (1 - \frac{a^2}{b^2})}{-ae}$$

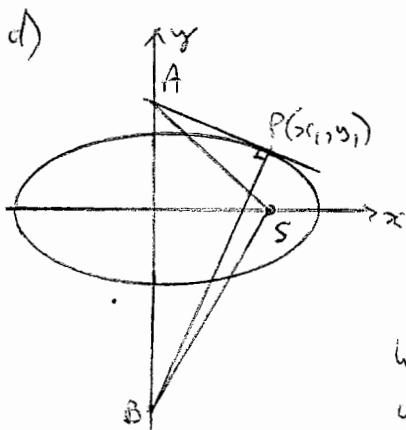
Now slope AS $\times$ slope BS

$$= \frac{b_1}{y_1 \times -ae} \times \frac{y_1 (1 - \frac{a^2}{b^2})}{-ae}$$

$$= \frac{b^2 - a^2}{a^2 e^2}$$

$$= -1 \text{ as } a^2 e^2 = a^2 - b^2$$

$$\therefore AS \perp BS$$



When $x = 0$, $y = \frac{b^2}{y_1}$

Normal at P,

$$\frac{x_1 y}{a^2} - \frac{x y_1}{b^2} = \frac{x_1 y_1}{a^2} - \frac{x_1 y_1}{b^2}$$

When $x = 0$,

$$y = \frac{a^2}{x_1} \left(\frac{x_1 y_1}{a^2} - \frac{x_1 y_1}{b^2} \right)$$

$$= y_1 \left(1 - \frac{a^2}{b^2} \right)$$

tangent at P,

$$\frac{x x_1}{a^2} + \frac{y y_1}{b^2} = 1$$